NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Computer Science & Information Technology

Intern Management Portal

with Timetable Scheduling and Attendance Tracking

Course Title	Web Engineering
Course Code	CT-362
Semester	Spring 2026
Submission Date	May 4th, 2026

Submitted By

Group Member Name	Roll Number
Ahmad Raza	CT-23086
Haiqa Anis Khan	CT-23060
Emman Nadeem	CT-23072
Zehra Saleem	CT-23069

Submitted To:

Sir Wajhi Uddin

Table of Contents

Abstract

This project is an Intern Management Portal designed for managing interns, projects, tasks, timetables, meetings, and attendance in a structured digital system. The portal provides separate dashboards for Admin, Student, and Team Lead users. Admins can manage interns, assign projects, create tasks, schedule meetings, and view attendance reports. Students can view their assigned tasks, update progress, manage their timetable availability, and mark attendance during scheduled meeting slots. Team Leads can view assigned interns, manage tasks, and monitor timetable availability.

The system uses a MERN-based architecture with React and TypeScript on the frontend, Node.js and Express.js on the backend, and MongoDB as the database. The attendance module includes time-based attendance marking and location-based validation using geolocation. The timetable module follows academic calendar logic and supports projected weekly availability and meeting schedules.

1. Introduction

Internship programs require proper coordination between students, administrators, and team leads. In many departments, intern data, task allocation, attendance records, and scheduling are managed manually or through scattered tools. This can create confusion, delays, missed updates, and difficulty in tracking student progress.

The Intern Management Portal solves this problem by providing a centralized web-based platform. It allows the department to manage intern records, assign projects and tasks, schedule meetings according to student availability, and track attendance digitally. The portal improves communication, reduces manual work, and gives each user role a dedicated dashboard.

2. Problem Statement

The existing manual process of managing interns is time-consuming and difficult to monitor. Admins may need to track intern details, task assignments, project allocation, attendance, and timetable availability separately. Students also face difficulty in knowing their assigned tasks, deadlines, meetings, and attendance status in one place.

Therefore, there is a need for a centralized system that allows admins, students, and team leads to manage internship-related activities efficiently. The system should support role-based access, task management, project management, timetable scheduling, and attendance tracking.

3. Objectives

The main objectives of this project are:

- To create a centralized portal for internship management.
- To provide separate dashboards for Admin, Student, and Team Lead.
- To allow admins to view, search, and remove interns.
- To allow admins to create and manage projects.
- To allow admins and team leads to assign and monitor tasks.
- To allow students to update task progress and request approvals.
- To allow students to mark free/busy timetable slots.
- To allow admins to schedule meetings based on student availability.
- To implement attendance marking based on scheduled meetings.

- To provide attendance reports for admin monitoring.

4. Scope of the Project

The scope of this project includes user authentication, intern profile management, project management, task management, timetable scheduling, attendance tracking, and reporting. The system supports three major user roles: Admin, Student, and Team Lead.

The admin has control over intern records, projects, tasks, scheduling, and reports. Students can view their dashboard, update their profile, manage timetable availability, view assigned tasks, and mark attendance. Team Leads can view assigned interns, manage team tasks, and check timetable availability.

The system is designed for academic internship management and can be extended in the future for larger organizations or industrial internship programs.

5. Tools and Technologies Used

5.1 Frontend

The frontend of the system is developed using React.js with TypeScript. Tailwind CSS and shadcn UI components are used for styling and reusable interface elements. React Router is used for page navigation, while Axios is used for API communication with the backend.

5.2 Backend

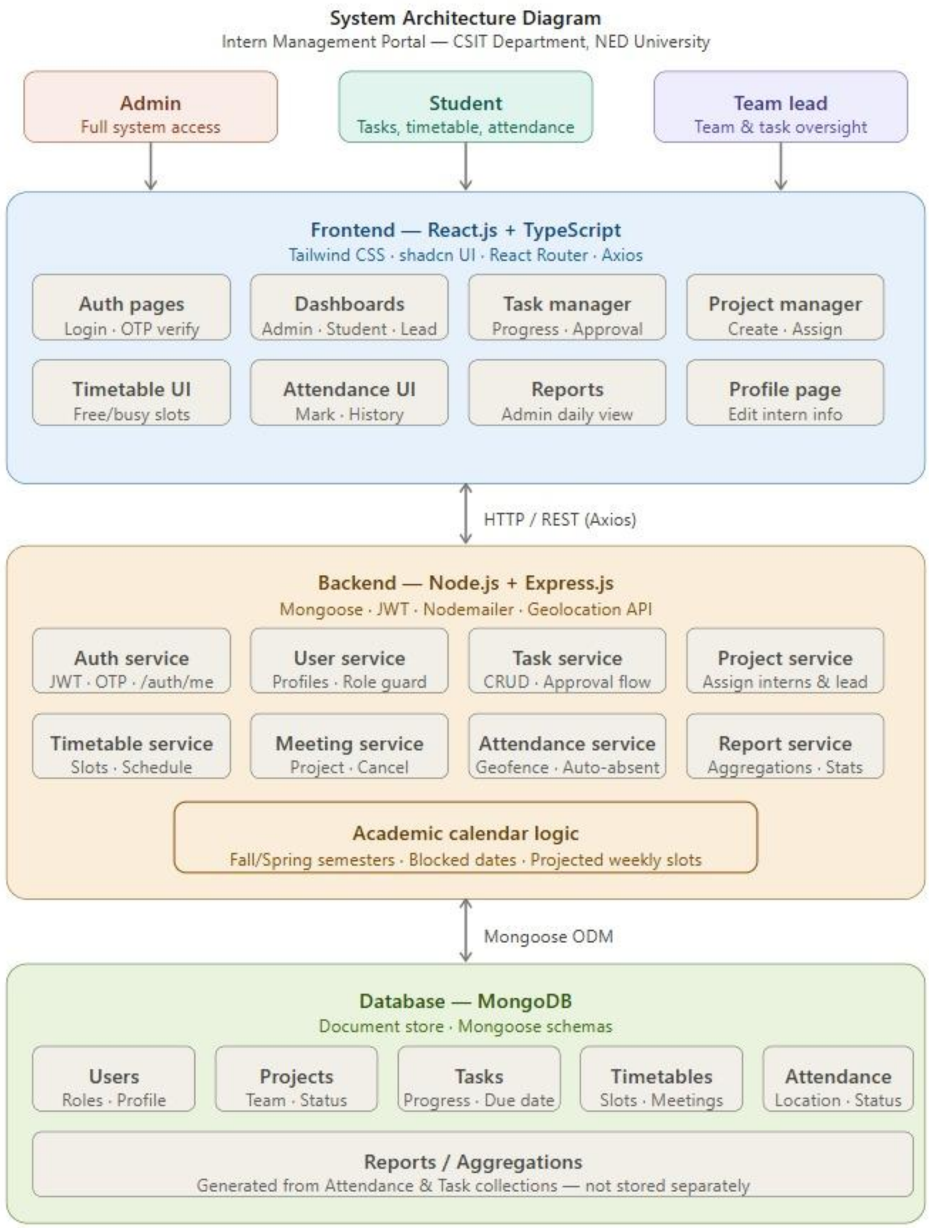
The backend is developed using Node.js and Express.js. MongoDB is used as the database, and Mongoose is used for schema design and database operations. JSON Web Token is used for authentication. Nodemailer is used for OTP-based email verification. The system also uses browser geolocation for attendance marking.

Layer	Technology	Purpose
Frontend	React.js + TypeScript	UI Development
Frontend	Tailwind CSS + shadcn UI	Styling & Components
Frontend	React Router	Page Navigation
Frontend	Axios	API Communication
Backend	Node.js + Express.js	Server & REST APIs
Backend	MongoDB + Mongoose	Database & ORM
Backend	JWT	Authentication
Backend	Nodemailer	Email / OTP Verification
Backend	Geolocation API	Attendance Location Check

6. System Architecture

The system follows a client-server architecture. The frontend acts as the client-side application where users interact with the portal. The backend handles API requests, authentication, business logic, validation, and database operations. MongoDB stores all user, project, task, timetable, and attendance data.

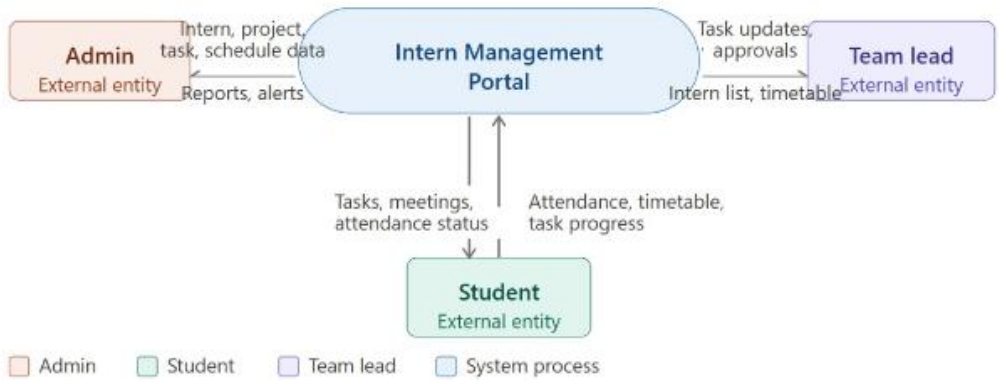
The frontend sends requests to the backend through API endpoints. The backend verifies the user token, checks the user role, performs the requested operation, and returns the response to the frontend. This separation makes the system easier to maintain and extend.



System Architecture of Intern Management Portal

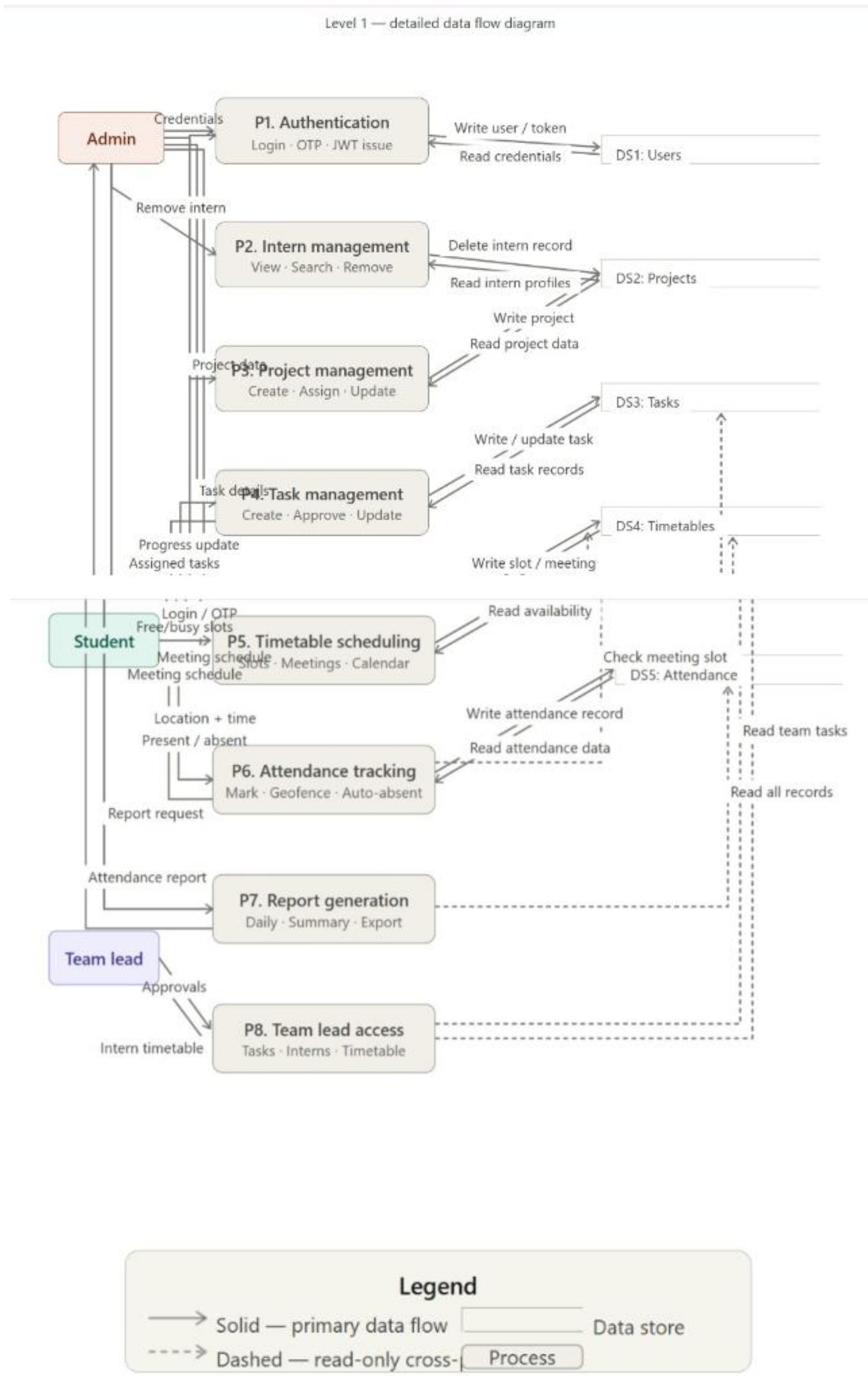
7. Data Flow Diagram

7.1 DFD Level 0



DFD LEVEL 0 CONTEXT DIAGRAM

7.2 DFD Level 1



8. User Roles

8.1 Admin

The Admin is responsible for managing the overall internship program. The admin can view interns, remove interns, manage projects, assign tasks, schedule meetings, view timetable availability, and generate attendance reports.

8.2 Student

The Student can log in to the portal, view assigned tasks, update task progress, request approval, manage timetable availability, view scheduled meetings, and mark attendance during meeting slots.

8.3 Team Lead

The Team Lead can view interns assigned under their projects, manage tasks related to their team, approve or reject task progress, and view student timetable availability.

9. Authentication and User Management

The system includes authentication using email and password. It also supports OTP-based verification during registration. Users must verify their account before using the portal. The backend uses JWT tokens to protect private routes and identify the logged-in user.

The user model stores basic information such as name, email, password, role, profile picture, discipline, batch, roll number, phone number, semester, and date of joining. The /auth/me route is used by the frontend to load the currently logged-in user.

10. Admin Dashboard Module

The Admin Dashboard gives the admin a quick overview of the internship program. It displays dynamic information such as total interns, active projects, pending tasks, and meetings. The admin can quickly navigate to intern management, project management, task management, attendance reports, and timetable scheduling.

This dashboard helps the admin monitor the overall status of the internship program from one place.

11. Intern Management Module

The Intern Management module allows the admin to view all registered students with the role of intern/student. The admin can search interns by name, email, discipline, semester, roll number, or batch. The admin can also open a read-only profile view of each intern.

The remove intern feature deletes the selected intern from the database and also removes related records such as tasks, projects, timetable data, and attendance records. This helps keep the database clean when an intern is removed from the system.

12. Project Management Module

The Project Management module allows admins to create, update, delete, and assign projects. Each project has a title, description, status, color, due date, assigned interns, and a team lead. Admins can assign students to projects and also select one student as the team lead.

This module helps organize internship work into structured projects and ensures that each intern is connected to the correct project and team lead.

13. Task Management Module

The Task Management module allows admins and team leads to create and manage tasks. A task contains a title, description, subheading, assigned students, assigned by user, project reference, due date, status, progress, start date, and completion date.

Students can update their task progress and request approval. Team leads and admins can approve, reject, reset, or update task statuses. The system also supports missed task handling when deadlines pass. This module helps track intern work progress and ensures proper approval flow.

14. Timetable and Scheduling Module

The timetable module allows students to mark their availability as free or busy. The timetable uses weekdays from Monday to Friday and fixed time slots from 8:30 AM to 4:30 PM.

The system follows academic calendar logic. It supports Fall and Spring semesters starting from 2026 onward. Certain months such as January, June, August, and December are kept open according to the project requirements. Dates outside the academic calendar are blocked.

When a student marks a slot free or busy, the system projects that availability to the same weekday across the semester. For example, if a student marks Monday 8:30-9:20 as free, all Mondays of that academic period can follow the same availability unless the student changes it.

Admins can schedule meetings based on available student slots. Meeting scheduling is projected across the same weekday in the academic period. However, meeting cancellation is date-specific, meaning the admin can cancel a meeting for one specific date without removing the meetings from other weeks.

15. Attendance Management Module

The attendance module is connected with the timetable and meeting system. Attendance is only available for scheduled meeting slots. Students can mark attendance only during the actual meeting time.

The system checks the current date and time according to the configured timezone. It also uses browser geolocation to capture the student's latitude, longitude, and accuracy. The backend calculates the distance between the student's location and the allowed attendance location. If the student is within the configured geofence radius, attendance can be marked as present.

If a meeting slot ends and the student has not marked attendance, the system can mark the record as absent. This makes attendance tracking automatic and more reliable.

16. Student Attendance Page

The Student Attendance page shows the student's meetings for the current day. It displays whether the meeting is upcoming, open for attendance, present, or absent. The student can only click the attendance button when the meeting status is open.

The page also shows attendance history and calculates summary values such as present count, absent count, and attendance percentage. This helps students monitor their own attendance record.

17. Admin Attendance Reports Module

The Attendance Reports page allows the admin to view attendance records for selected academic dates. The admin can navigate by month and week, select a date, and generate a daily attendance report.

The report includes total meetings, present students, absent students, open meetings, upcoming meetings, and attendance rate. It also displays individual student records with meeting time, status, marked time, and distance information.

This module helps the admin monitor daily attendance performance and identify absent students easily.

18. Team Lead Module

The Team Lead module gives team leads access to interns assigned under their projects. Team leads can view their dashboard, manage assigned tasks, approve or reject progress, and view timetable availability of their assigned interns.

The team lead timetable page shows availability and scheduled meeting information for only those interns who belong to the team lead's projects. This keeps access limited and role-specific.

19. Student Dashboard and Profile Module

The Student Dashboard provides students with a simple view of their internship activities. Students can access tasks, attendance, timetable, and profile pages.

The Student Profile page allows students to view and update their profile information such as name, discipline, batch, roll number, phone number, semester, and date of joining. This keeps intern records updated in the system.

20. Database Design

The system uses MongoDB collections for storing data. The main models are described below:

Collection	Description
User	Stores student, admin, and team lead account details including name, email, password, role, profile picture, discipline, batch, roll number, phone number, semester, and date of joining.

Collection	Description
Project	Stores project title, description, status, color, team lead reference, assigned interns, and due date.
Task	Stores task details, assigned users, assigned by user, project reference, due date, status, progress, and approval information.
Timetable	Stores each student's availability slots with day, time, date, status, label, meeting date, and assigned by user reference.
Attendance	Stores attendance records linked to student, timetable slot, date, meeting title, status, marked time, location coordinates, accuracy, distance, and geofence result.

21. API Design

The backend provides REST APIs for communication between frontend and database. Important API groups include:

Authentication APIs for login, registration, OTP verification, and current user profile.
User APIs for fetching students, viewing profiles, and removing interns.
Project APIs for creating, updating, deleting, and assigning projects.
Task APIs for creating tasks, updating progress, approving tasks, and deleting tasks.
Timetable APIs for saving student availability, fetching student timetables, admin scheduling, and team lead timetable views.
Attendance APIs for today's meetings, marking attendance, attendance history, and admin daily reports.

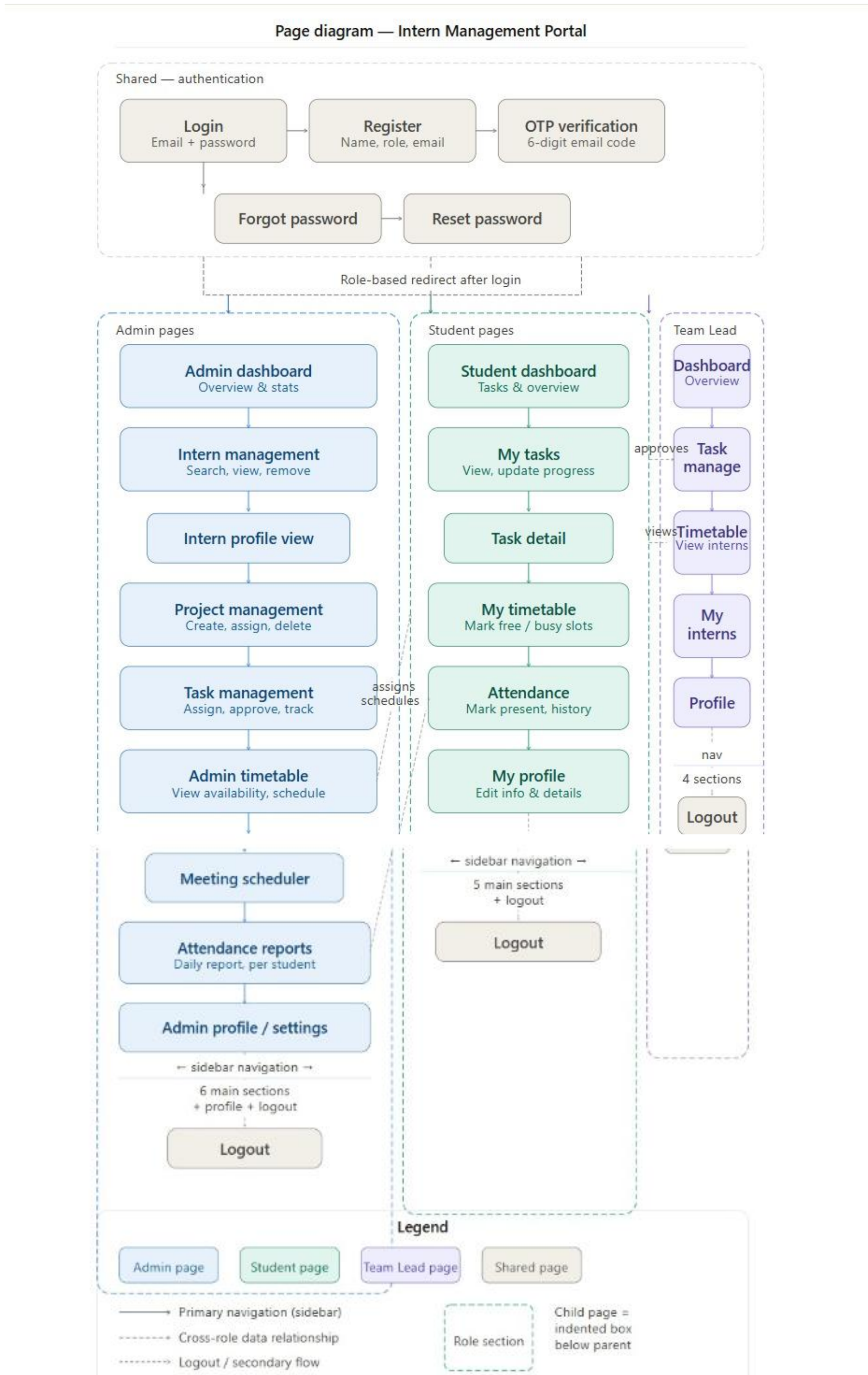
These APIs allow the frontend to interact with the backend in a structured and secure way.

22. Frontend Design

The frontend is developed using React.js and TypeScript. It uses a clean dashboard-based interface with separate layouts for Admin, Student, and Team Lead. Shared navbar and sidebar components are used to keep the design consistent across pages.

The UI uses cards, tables, badges, buttons, dialogs, and forms. Tailwind CSS and shadcn UI are used to create a professional and responsive interface. The design uses color-coded statuses such as green for free/present, gray for busy, blue for task/open, purple for meeting, and red for absent.

22.1 Page Diagram



23. Backend Design

The backend is developed using Express.js. It handles authentication, route protection, role-based access, database operations, and business logic. Mongoose models are used to define database schemas.

The backend validates important operations such as protected routes, academic timetable dates, task conflicts, meeting scheduling, attendance time windows, and geofence-based attendance marking. This ensures that the system works reliably and prevents invalid data from being saved.

24. Testing and Validation

Testing was performed by checking the system from Admin, Student, and Team Lead perspectives.

Admin Testing
Intern management, project creation, task assignment, timetable viewing, meeting scheduling, meeting cancellation, and attendance report generation were checked and verified.

Student Testing
Login, profile update, timetable availability marking, task progress update, attendance marking, and attendance history were checked and verified.

Team Lead Testing
Assigned intern visibility, task management, approval flow, and timetable viewing were checked and verified.

The testing confirmed that role-based functionality works correctly and that timetable, meeting, and attendance modules are connected properly.

25. Results and Output

The final system successfully provides a working Intern Management Portal with multiple role-based dashboards. Admins can manage interns, projects, tasks, timetables, meetings, and attendance reports. Students can manage their timetable, view tasks, update progress, and mark attendance. Team Leads can monitor assigned interns and manage team tasks.

The attendance module successfully connects scheduled meetings with time-based and location-based attendance marking. The timetable module supports academic calendar logic and projected weekly scheduling.

26. Limitations

Although the system is functional, some limitations still exist:

- The system currently depends on browser geolocation, which may vary depending on device accuracy and permission settings.
- The attendance system requires correct environment configuration for location and timezone.
- The project requires proper deployment configuration for frontend and backend communication.
- Some report pages can be further improved by adding charts, export options, and more advanced filtering.

27. Future Enhancements

Future improvements that can be incorporated into the system include:

- PDF export for attendance reports.
- Email notifications for scheduled meetings and pending tasks.
- More advanced analytics for intern performance.
- Calendar integration with Google Calendar.
- Real-time notifications using WebSockets.
- Admin-controlled semester calendar settings.
- Better mobile responsiveness and PWA support.
- More detailed team lead performance reports.
- File upload support for task submissions.

28. Conclusion

The Intern Management Portal provides a complete digital solution for managing interns in an academic department. It reduces manual work and improves coordination between admins, students, and team leads. The system successfully combines intern records, project management, task tracking, timetable scheduling, and attendance monitoring into one platform.

By using React, Node.js, Express, and MongoDB, the portal is scalable and maintainable. The role-based structure, academic timetable logic, and attendance tracking make it suitable for real internship management needs.

29. References

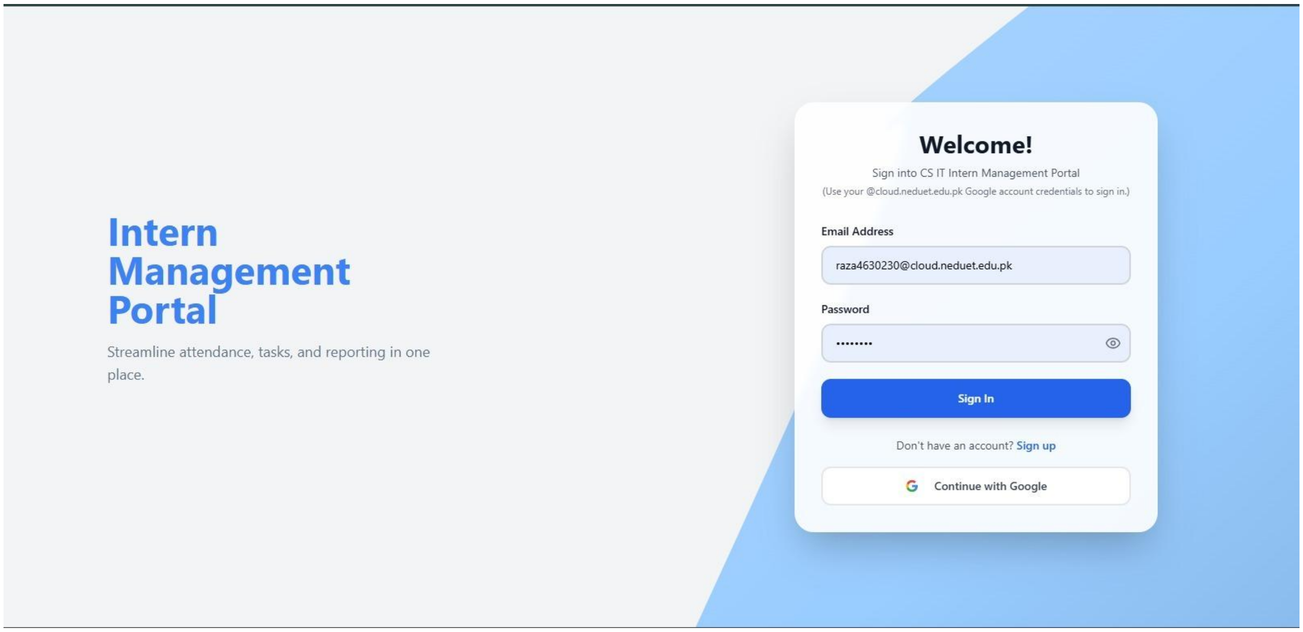
- React.js Official Documentation — https://reactjs.org/docs
- Node.js Official Documentation — https://nodejs.org/en/docs
- Express.js Official Documentation — https://expressjs.com
- MongoDB Official Documentation — https://www.mongodb.com/docs
- Mongoose Official Documentation — https://mongoosejs.com/docs
- Tailwind CSS Official Documentation — https://tailwindcss.com/docs
- shadcn/ui Official Documentation — https://ui.shadcn.com/docs
- MDN Web Docs - Geolocation API — https://developer.mozilla.org/en-US/docs/Web/API/Geolocation_API

30. Appendix

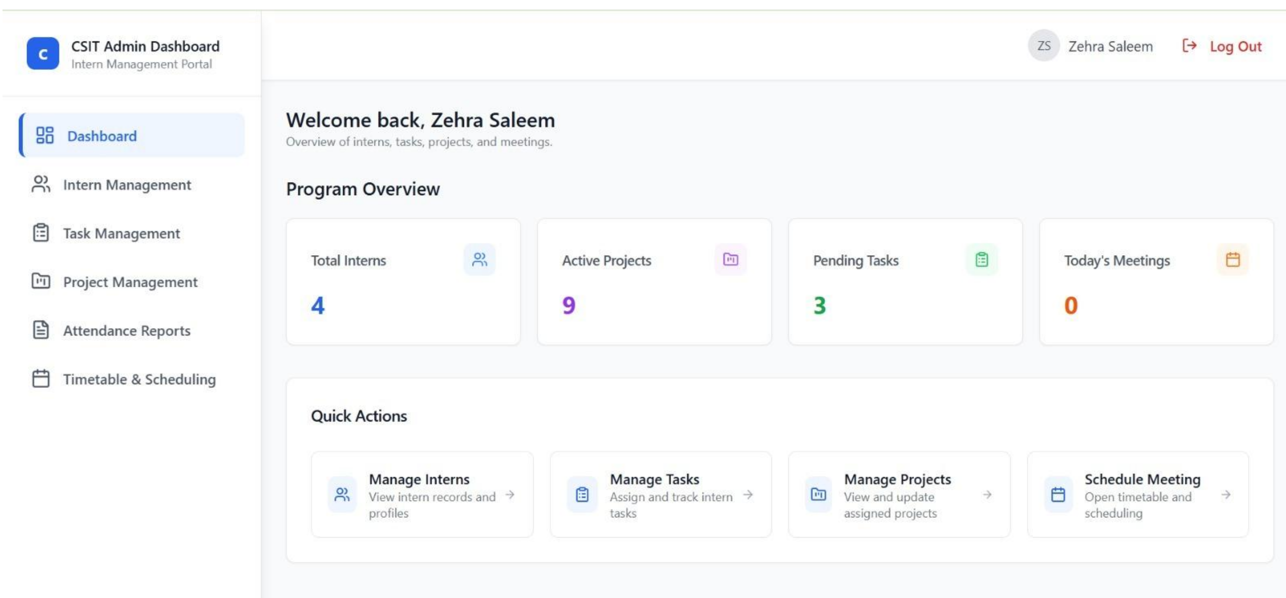
The following screenshots and materials should be included in the final submission:

A. Screenshots

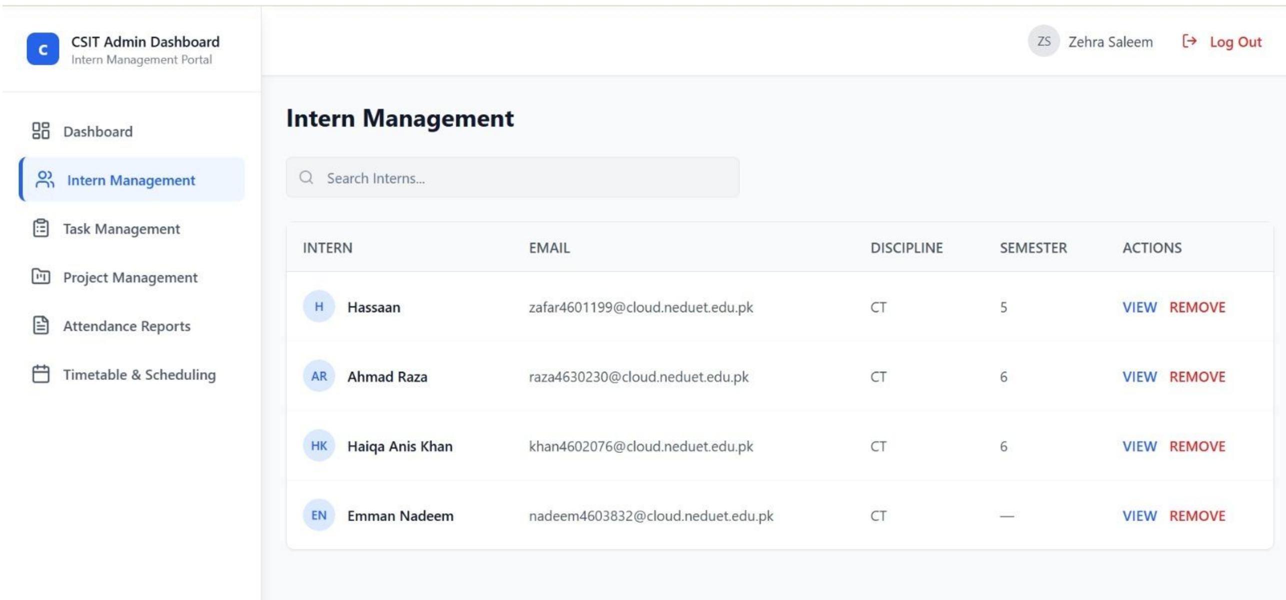
- Login page



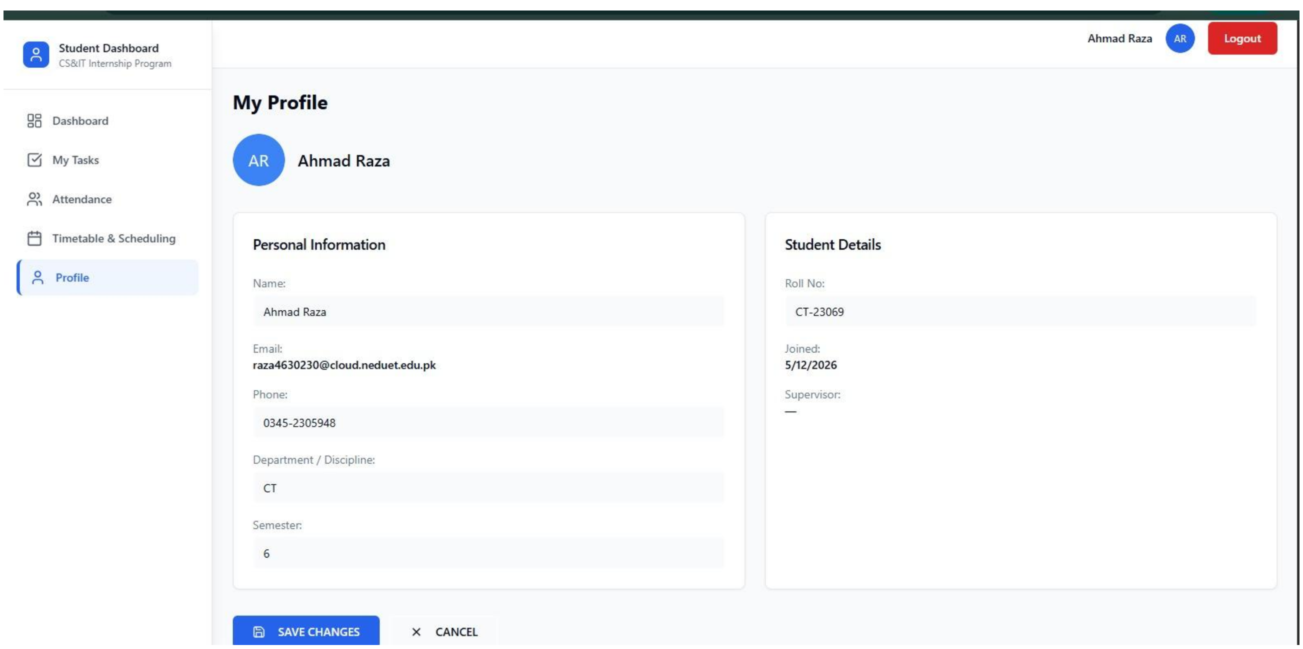
- Admin dashboard



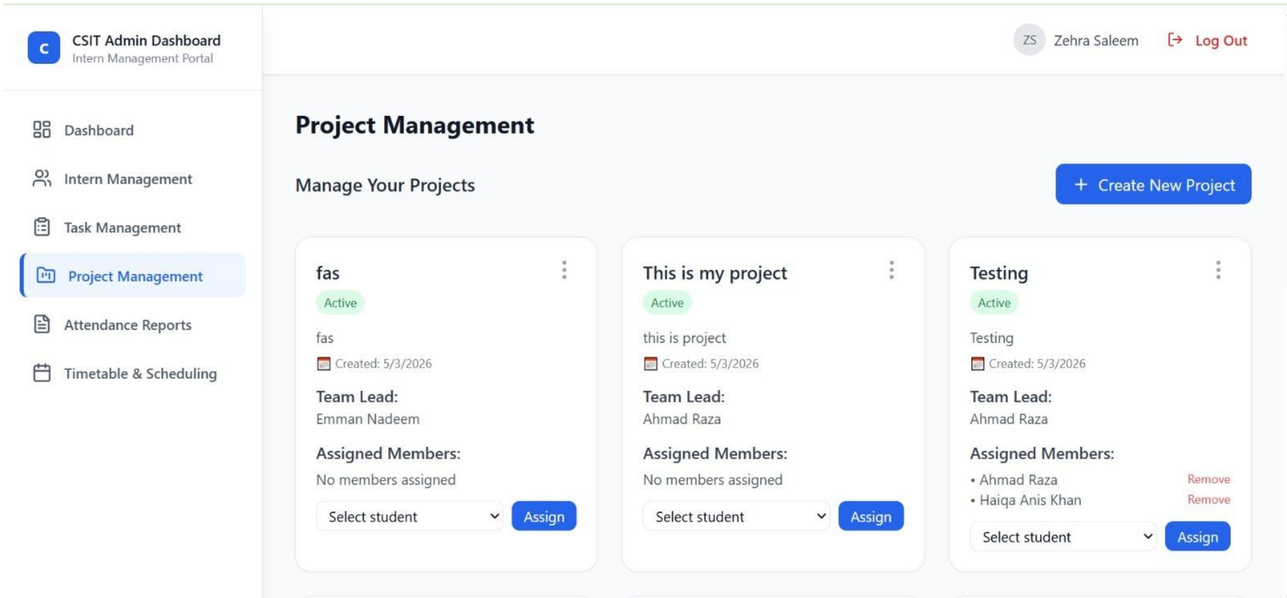
- Intern management page



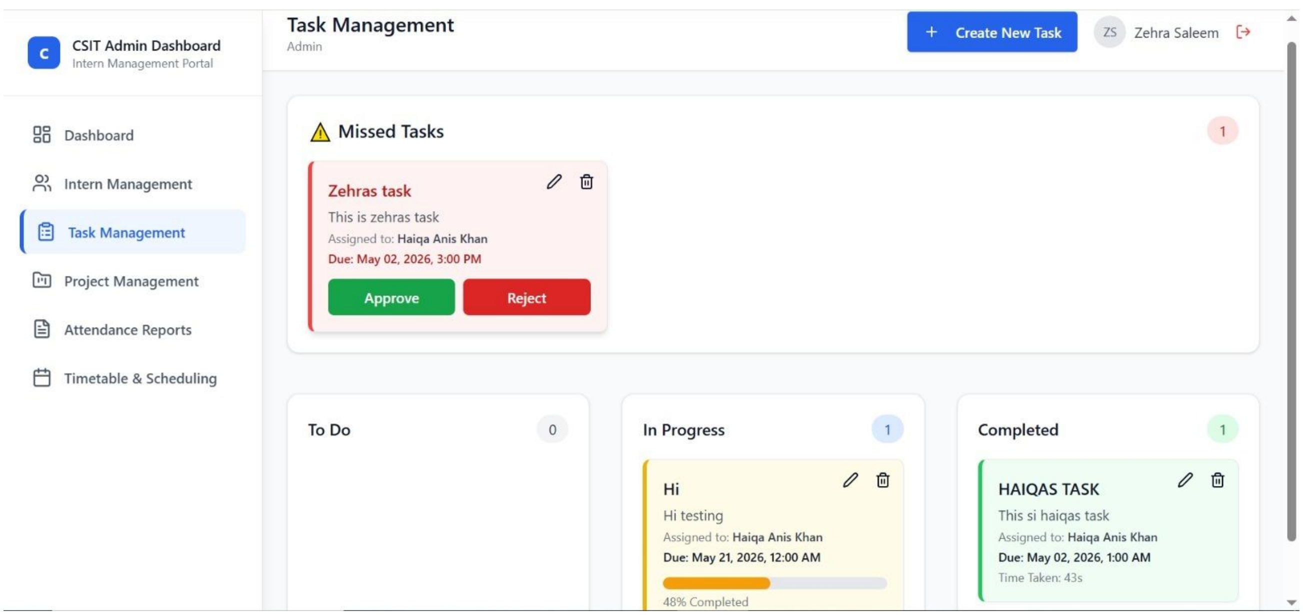
- Intern profile view



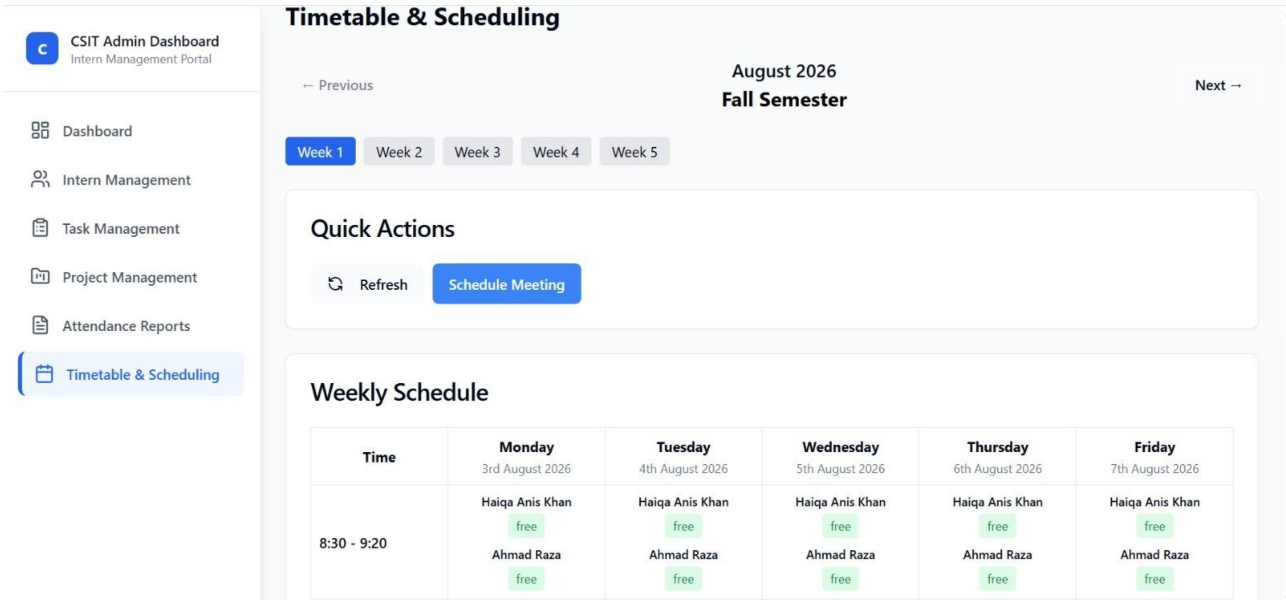
- Project management page



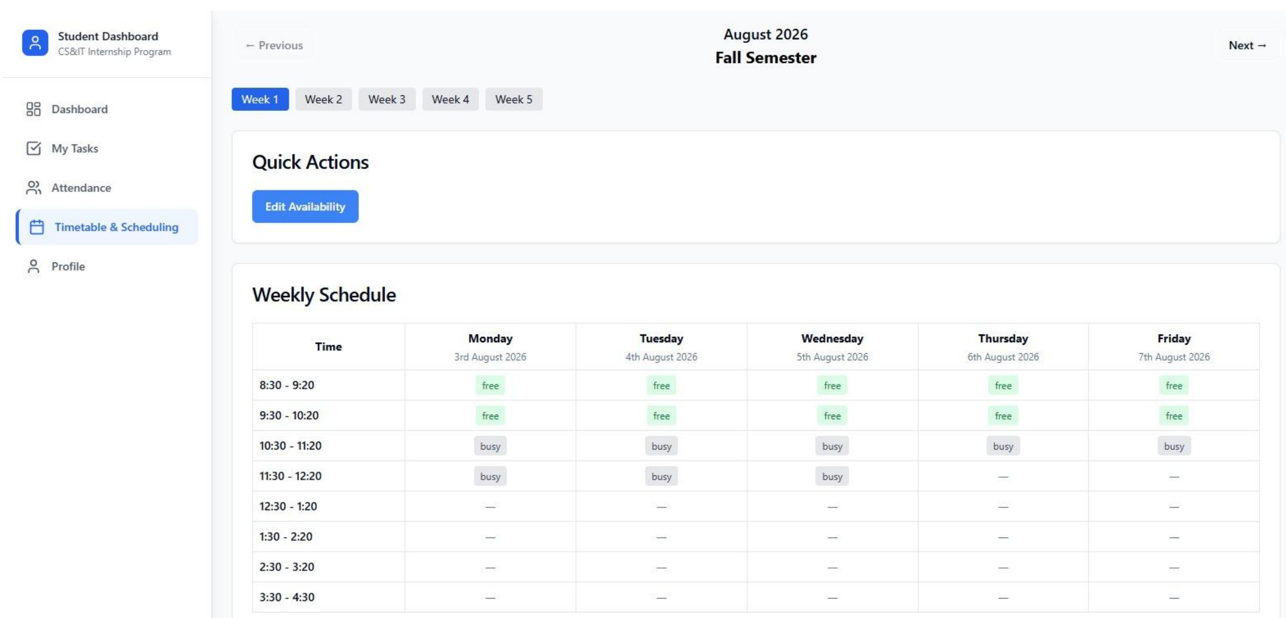
- Task management page



- Admin timetable page

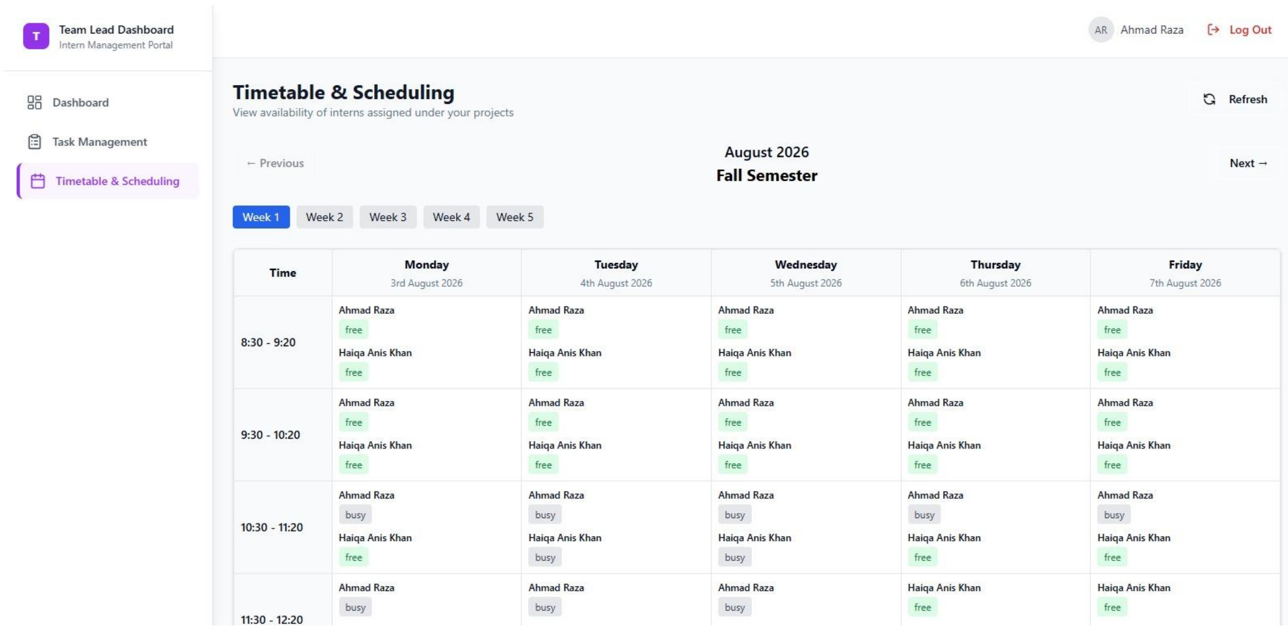


- Student timetable page

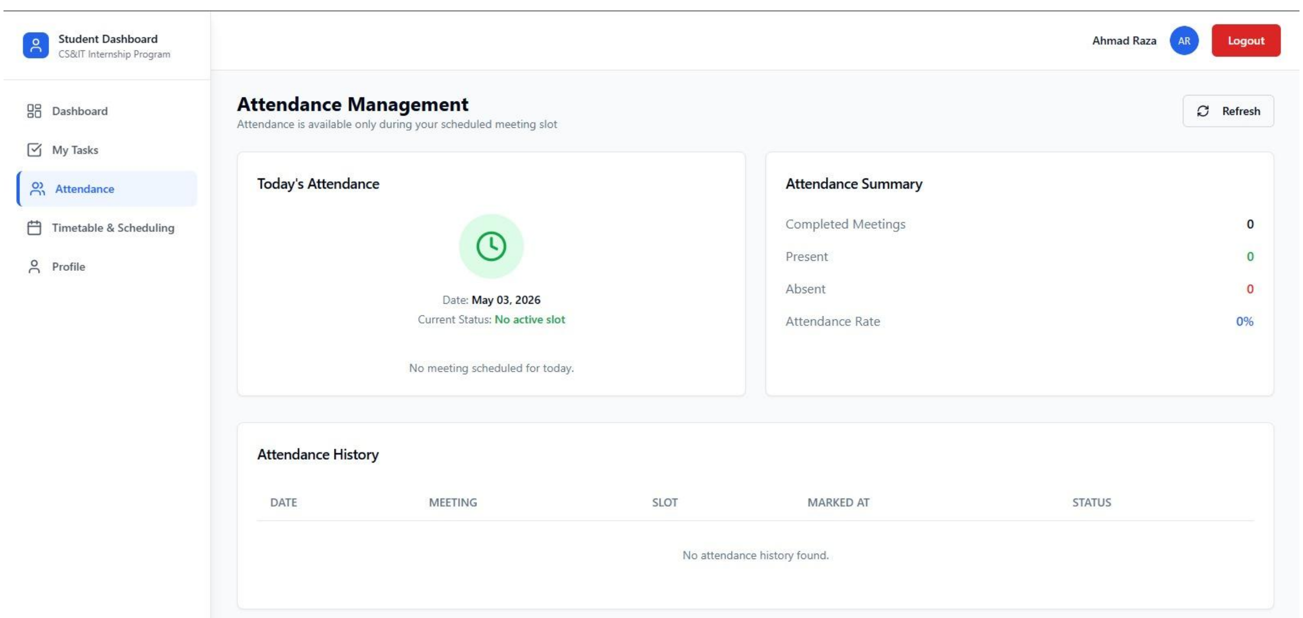


Time	Monday 3rd August 2026	Tuesday 4th August 2026	Wednesday 5th August 2026	Thursday 6th August 2026	Friday 7th August 2026
8:30 - 9:20	free	free	free	free	free
9:30 - 10:20	free	free	free	free	free
10:30 - 11:20	busy	busy	busy	busy	busy
11:30 - 12:20	busy	busy	busy	—	—
12:30 - 1:20	—	—	—	—	—
1:30 - 2:20	—	—	—	—	—
2:30 - 3:20	—	—	—	—	—
3:30 - 4:30	—	—	—	—	—

- Team lead timetable page



- Student attendance page



- Admin attendance reports page